

Submission dossier — pulse oximeter

Fixture pulse-oximeter-literal-complete · expected overall result: PASS

SYNTHETIC TEST DOCUMENT — assembled from the MedDevAudit-IN evaluation fixtures for the purpose of exercising the auditor. Not a real submission. Device names, companies, certificate and licence numbers are fictitious.

2.1 Import licence. The device is imported under import licence Form MD-15 number IMP/MD/2024/004518, granted by the Central Licensing Authority, CDSCO, on 14 March 2024 and valid until 13 March 2029. A copy of the licence is placed at Annex 2. The licence number is reproduced verbatim on the carton label (Annex 8) and on the Bill of Entry used for customs clearance, so that the three documents can be reconciled at the port.

2.2 Application and authorised agent. The application was made in Form MD-14 dated 02 January 2024 by Sundaram Healthcare Distributors Pvt Ltd, 41 Anna Salai, Chennai 600002, Tamil Nadu, India, holder of wholesale licence 20B/21B No. TN/CH/2019/3312, acting as the Indian authorised agent of Meridian Medical GmbH. The notarised power of attorney dated 18 December 2023 by which Meridian Medical GmbH appoints Sundaram Healthcare Distributors Pvt Ltd for the purposes of Rule 35 is at Annex 3.

2.3 Free Sale Certificate. A Free Sale Certificate for the FS-200 Fingertip Pulse Oximeter was issued by the Bundesinstitut für Arzneimittel und Medizinprodukte (BfArM), the national regulatory authority of Germany, the country of origin, under certificate number 2023/FSC/88214 dated 05 November 2023, confirming that the device is freely sold in Germany. The certificate is valid for two years and is placed at Annex 4 with an apostille.

3.1 Quality management system. The manufacturing site at Industriestrasse 12, 82256 Fürstentfeldbruck, Germany, holds ISO 13485:2016 certificate number DE-QMS-2022-77413 issued by TÜV SÜD Product Service GmbH. Scope of certification: design, development and manufacture of non-invasive patient monitoring equipment including pulse oximeters. Date of issue 11 April 2022; valid until 10 April 2026. The certificate and the most recent surveillance audit report (October 2024) are at Annex 5.

4.1 Risk management. Risk management has been carried out in accordance with ISO 14971:2019 across the device lifecycle. The risk management file at Annex 6 comprises the risk management plan RMP-FS200-01, the hazard identification and risk analysis worksheet (design FMEA, 64 identified hazards), the risk evaluation and risk control table with verification of effectiveness for each control, the residual risk evaluation, and the signed risk management report RMR-FS200-03, which concludes that the overall residual risk is acceptable in relation to the clinical benefit.

5.1 Electrical safety. Conformity with IEC 60601-1:2005 + A1:2012 + A2:2020 (general requirements for basic safety and essential performance) is demonstrated by test report number TR-60601-1-22884 issued by TÜV SÜD Product Service GmbH, an ILAC-MRA accredited test laboratory, dated 19 September 2023. The applied part is classified Type BF. The report covers dielectric strength, leakage currents in normal and single-fault condition, mechanical strength and marking, and concludes PASS on all applicable clauses. Full report at Annex 7.

5.2 Electromagnetic compatibility. Conformity with IEC 60601-1-2:2014 + A1:2020 is demonstrated by EMC test report number TR-EMC-22891 (TÜV SÜD Product Service GmbH, 22 September 2023). Emissions were tested to CISPR 11 Group 1 Class B. Immunity was tested at the levels specified for the home healthcare environment, including ESD ± 8 kV contact / ± 15 kV air, radiated RF 10 V/m, and proximity fields from wireless communications equipment. The declared electromagnetic environment is home healthcare. The manufacturer's EMC declaration tables are reproduced in section 8 of the Instructions for Use. Report at Annex 7.

8.1 Label — manufacturer. The device label and the outer carton artwork (Annex 8, drawings LBL-FS200-DEV-03 and LBL-FS200-CTN-03) carry the manufacturer symbol followed by: "Manufactured by:

Meridian Medical GmbH, Industriestrasse 12, 82256 Fürstenfeldbruck, Germany". This is the address of the manufacturing site named on the ISO 13485 certificate, not a sales office.

8.2 Label — importer and licence. The carton artwork (Annex 8) carries, in addition to the manufacturer block: "Imported and marketed by: Sundaram Healthcare Distributors Pvt Ltd, 41 Anna Salai, Chennai 600002, India" and "Import licence number: IMP/MD/2024/004518". The licence number matches the MD-15 licence at Annex 2.

8.3 Label — language. The FS-200 is intended for home use. The carton artwork (Annex 8) carries the statement "For Sale in India" in English with the Hindi (Devanagari) rendering "इसका उपयोग भारत में ही करना चाहिए" and the essential particulars — device name, intended use, manufacturer and importer — are presented bilingually in Hindi and English in the same panel.

8.4 Label — symbols. Symbols used on the artwork conform to ISO 15223-1:2021. The symbol glossary reproduced in section 1 of the Instructions for Use lists each symbol with its meaning: manufacturer, date of manufacture, batch code (LOT), serial number (SN), catalogue number (REF), consult instructions for use, Type BF applied part, temperature and humidity limits, and "keep dry". Annex 8 marks the position of each symbol on the device label and carton.

9. Instructions for Use. The IFU (document IFU-FS200-EN rev 6, Annex 9) is supplied in English with every unit and contains: intended use and intended user; indications for use; step-by-step operating instructions; contraindications; warnings, cautions and precautions; the residual risks communicated to the user; cleaning and maintenance instructions; storage and transport conditions ("20 to +60 °C, 10 to 93 % RH); expected service life of five years; and the contact details for reporting an adverse event or complaint in India (Materiovigilance cell, Sundaram Healthcare Distributors Pvt Ltd, vigilance@sundaramhealth.in, +91 44 4000 1200).

1. Device Master File — index. This Device Master File is submitted in the format of the Fourth Schedule, Part II. Table of contents: 1 executive summary; 2 device description and intended use; 3 manufacturing process and site details; 4 list of applied standards; 5 design verification and validation data; 6 clinical accuracy study; 7 stability and shelf-life data; 8 labelling and packaging set; 9 Instructions for Use; 10 post-market surveillance plan and vigilance procedure. Each numbered section is present in this bundle and every annex referenced in the sections is attached (Annexes 1–12).

6.1 Clinical accuracy (SpO₂). A controlled desaturation study per ISO 80601-2-61:2017 clause 201.12.1.101 was conducted at the University of California San Francisco Hypoxia Laboratory in 12 healthy adult volunteers, yielding 216 paired data points against a Radiometer ABL90 CO-oximeter over the saturation range 70 to 100 %. The cohort comprised 4 subjects with dark (Fitzpatrick V–VI), 4 intermediate and 4 light pigmentation. Result: SpO₂ accuracy root mean square (ARMS) 1.9 % over 70–100 %. Bland-Altman analysis (Annex 6A): mean bias " 0.3 %, limits of agreement " 4.1 to +3.5 %.

8.6 Label and IFU — declared performance. The label and IFU section 9 state: SpO₂ accuracy ARMS 1.9 % over the saturation range 70–100 %; pulse rate range 30–250 bpm with pulse rate accuracy ± 2 bpm; emitter wavelengths 660 nm and 905 nm with maximum optical output power below 15 mW; and the list of compatible sensors validated with this monitor — FS-ADT-01 (adult) and FS-PED-01 (paediatric). The IFU states that the ARMS figure is a statistical measure and is not the measurement uncertainty of an individual reading.

5.4 Alarm system. The FS-200 provides physiological alarms (low SpO₂ — high priority; low and high pulse rate — medium priority) and technical alarms (low battery, probe off — low priority). Conformity with IEC 60601-1-8:2006 + A1:2012 + A2:2020 is demonstrated by test report TR-ALM-22903 (TÜV SÜD, Annex 7C), which verifies alarm condition priorities, audible alarm signal characteristics (bursts, pitch, sound pressure) and visual alarm signal characteristics, alarm limit setting range and default limits, the alarm-off and audio-paused indications, and an alarm system delay below 10 s.

7. Software. The embedded firmware is developed under a lifecycle process conforming to IEC

62304:2006 + A1:2015. Software safety classification: Class B, justified because a software failure could contribute to a hazardous situation (delayed or erroneous saturation reading) leading to non-serious injury. The documentation set (Annex 10) comprises the software development plan SDP-FS200, software requirements specification SRS-FS200 rev 4, software architecture document, unit and integration verification records, the software validation report SVVR-FS200, the known-anomaly list with residual anomaly justification, and the SOUP inventory (FreeRTOS 10.4.3, CMSIS-DSP 1.9.0) with its risk assessment.